# PHYS 7501 – Particles and Nuclei I – Syllabus

**Instructor**: Dr. Andrea Richard
**E-mail**: richarda1@ohio.edu
**Office**: 204 Edwards Accelerator Laboratory
**Office Hours**: Fridays at 10AM or By Appointment
**Class Location**: Clippinger 105
**Class Time**: Tuesdays/Thursdays 2 – 3:20PM

**Course Webpage:** https://github.com/arichard897/PHYS7501 (invitation only)

**Main Text:** *Modern Nuclear Chemistry*, Loveland, Morrissey, Seaborg (2nd Ed)
**Supplemental Texts**: *Introductory Nuclear Physics*, Wong (2nd Ed.) or *Introductory Nuclear Physics*, Krane

**Course Description**: The purpose of this course is to provide an overview of key topics in nuclear physics, primarily focusing on nuclear structure and nuclear reactions. Special topics such as nuclear astrophysics, nuclear applications, and experimental methods will also be briefly covered. No prior knowledge of nuclear physics will be assumed, and course content should be accessible to any graduate student in the Physics & Astronomy Department in their 2nd year or beyond.

The course will consist of lectures twice a week, intermixed with in-class group assignments. Additional learning tools will include homework assignments roughly every other week, a written midterm, and a final exam project/presentation.

**Homework & Group Work:** Students are encouraged to work together on homework assignments outside of class and are required to work together on group assignments during class. However, you must submit your own written and/or programmed solutions to each problem. Copying will not be tolerated and will result in a zero on the assignment and possibly in the course.

Exceptions for homework deadlines must be pre-arranged. Any in-class group work that is missed must be completed as if it were a homework assignment. Technically, only official university excuses will be accepted for absences, but other exceptions can be made with sufficient advance notice or for special circumstances.

**Exams:** There will be a midterm exam and a final exam. The midterm exam will be a written and graded exam during regular class time. The final exam will be an oral exam and written document, culminating in a presentation in front of the class and answering questions from the class and the instructor. Depending on the topic, this may necessarily draw on information from prior material in the class and your own reading material. Presentations will be peer-reviewed. A presentation rubric will be provided along with suggested topics.

**Attendance:** Attendance in this class is strongly encouraged. If you have a good reason not to be in class (e.g. personal issues, illness, unique opportunities like an academic conference or experiment off-campus), please let me know ahead of the class. For academic conferences or experiments, you must notify me in writing at least a week in advance. If you do not show up and do not provide an adequate reason, your grade will suffer. Any in-class assignments which are missed, will have to be made up as homework assignments.

**Academic Honesty:** I trust you will act in an academically honest fashion. If you have any questions about what does or does not constitute academic misconduct, please let me know.

**Academic misconduct** is a serious offense against the Student Code of Conduct (http://www.ohio.edu/communitystandards/academic/students.cfm) and can lead to sanctions up to and including expulsion from the university. The Student Code of Conduct defines academic misconduct to include plagiarism, cheating, submitting the same academic work more than once without consent of the instructor, fabrication (e.g., use of invented information or falsification of research or other findings), and misrepresenting facts to gain an academic advantage.

Plagiarism can be the most subtle of the offenses, and can happen inadvertently, through improper citation. Unintentional plagiarism occurs when material is copied or copied with minor changes without properly indicating it as another's work.

A student caught in a flagrant example of academic misconduct may be given a zero on the assignment or the course, and the case may be referred to the Office of Community Standards and Student Responsibility. The policies and procedures and sanctions which follow are detailed in the Student Code of Conduct. In addition, all cases of flagrant misconduct or suspected misconduct will be reported in writing to the Chair of the Department of Physics and Astronomy.

**Statement on Discrimination, Sexual Harassment and Bias:** The University endeavors to maintain an employment and educational environment that is free from discrimination and harassment. This includes discrimination and harassment on the basis of race, color, religion, age, ethnicity, national origin, national ancestry, sex, pregnancy, gender, gender identity or expression, sexual orientation, military service or veteran status, mental or physical disability, or genetic information. Faculty have a mandatory reporting responsibility to share information regarding sexual misconduct or information about a crime that may have occurred at Ohio University with the Office of University Equity and Civil Rights Compliance (ECRC). The ECRC office houses the university Title IX Coordinator, who can be reached by email at titleix@ohio.edu or by phone at 740-593-9140. In addition, students may contact the Survivor Advocacy Program (SAP) for confidential support. The SAP office is located in Lindley Hall 038. The SAP 24/7 Hotline is 740-597-7233. The SAP email is survivor.advocacy@ohio.edu

**Statement on Accessibility:** Any student who suspects he/she/they may need an accommodation based on the impact of a disability should contact the class instructor privately to discuss the student's specific needs and provide written documentation from the Office of Student Accessibility Services. If the student is not yet registered as a student with a disability, he/she/they should contact the Office of Student Accessibility Services.

**Statement on the Use of AI:** To be consistent with our scholarly values, students must cite any AI-generated material that informed their work and use quotation marks or other appropriate indicators of quoted material when appropriate. Students should indicate how AI tools informed their process and the final product, including what prompts you used and how you validated any AI-generated citations. Each assignment will provide additional guidance as to how these tools might be part of your process and how to be transparent about the use of AI in your work. Use of generative AI must be documented and cited appropriately.

**Protection of Intellectual Property**: The lectures, classroom activities, and all materials within this class and developed by the instructor are copyrighted in the name of Andrea Richard on August 26, 2025.

**Grading:** This class will follow a traditional grading scale for homework and exams but will make use of "ungrading" for self-assessment and reflection. Homework, exams, and in-class participation/group work are all equally weighted. We will individually check in at regular intervals to discuss how the course is going. Effort will be emphasized over achievement. I want you to enjoy what you're learning and work hard because you're enjoying it.

**PHYS 7501 – Particles and Nuclei I – Tentative Schedule**

*Subject to change. Advance notice will be given as early as possible.
*If I need to miss class due to conferences, experiments, etc. we will re-schedule or have a virtual session.

| Date | Chapters | Topic | Due |
|---|---|---|---|
| **August 26** | 1,2 | Nuclear Properties | |
| **August 28** | 1,2 | Nuclear Properties | |
| **September 2** | 1,2,5 | Nuclear Phenomenology | |
| **September 4** | 1,2,5 | Nuclear Phenomenology | |
| **September 9** | 6 | Nuclear Structure | HW #1 |
| **September 11** | 6 | Nuclear Structure | |
| **September 16** | 6 | Nuclear Structure | |
| **September 18** | 3 | Radioactive Decay | |
| **September 23** | 7 | Alpha Decay | HW #2 |
| **September 25** | 8 | Beta Decay | |
| **September 30** | 9 | Gamma Decay | |
| **October 2** | 11 | Fission | |
| **October 7** | **Midterm Exam** | | |
| **October 9** | 5,6 | Nuclear Forces | HW #3 |
| **October 14** | 10 | Nuclear Reactions | |
| **October 16** | 10 | Scattering | |
| **October 21** | 10 | Direct Reactions | |
| **October 23** | 10 | Resonant Reactions | |
| **October 28** | 10 | Statistical Reactions | HW #4 |
| **October 30** | 10 | Other Reaction Types | |
| **November 4** | 10,12 | Nuclear Astrophysics | |
| **November 6** | 12 | Nuclear Astrophysics | |
| **November 11** | Veteran's Day – No Class | | |
| **November 13** | 12 | Nuclear Astrophysics | HW #5 |
| **November 18** | 13,17,18 | Experimental Techniques | |
| **November 20** | 4 | Special Topics: Nuclear Medicine | |
| **November 25** | 15 | Special Topics: Nuclear Energy | |
| **November 27** | Thanksgiving Break – No Class | | |
| **December 2** | | Special Topics: TBD | HW#6 |
| **December 4** | | **Final Presentations Group 1** | |
| **December 9** | **Final Exam Presentation Group 2: 12:20PM** | | |